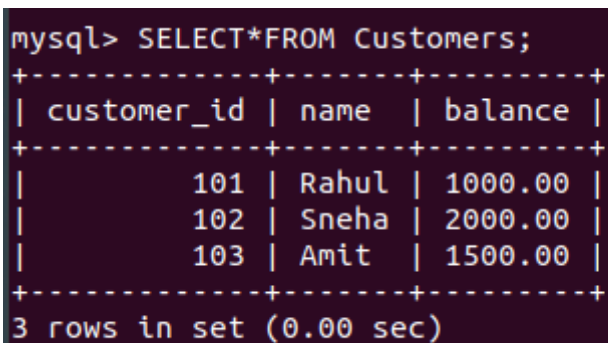


Practical No. 8

1.] Create tables and insert values:

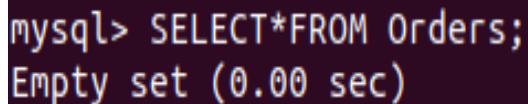
```
CREATE TABLE Customers (  
  customer_id INT PRIMARY KEY,  
  name VARCHAR(100),  
  balance DECIMAL(10,2)  
);
```

```
INSERT INTO Customers (customer_id, name, balance)  
VALUES  
(101, 'Rahul', 1000),  
(102, 'Sneha', 2000),  
(103, 'Amit', 1500);
```



```
mysql> SELECT*FROM Customers;  
+-----+-----+-----+  
| customer_id | name  | balance |  
+-----+-----+-----+  
|          101 | Rahul | 1000.00 |  
|          102 | Sneha | 2000.00 |  
|          103 | Amit  | 1500.00 |  
+-----+-----+-----+  
3 rows in set (0.00 sec)
```

```
CREATE TABLE Orders (  
  order_id INT PRIMARY KEY,  
  customer_id INT,  
  amount DECIMAL(10,2),  
  FOREIGN KEY (customer_id)  
  REFERENCES Customers(customer_id)  
  ON DELETE CASCADE  
);
```



```
mysql> SELECT*FROM Orders;  
Empty set (0.00 sec)
```

2.] Simple BEFORE INSERT TRIGGER :

```
DELIMITER $$  
CREATE TRIGGER simple_before_insert  
BEFORE INSERT ON Orders  
FOR EACH ROW
```

```
BEGIN
  IF NEW.amount < 0 THEN
    SIGNAL SQLSTATE '45000';
  END IF;
END$$
DELIMITER ;
```

3.] Simple AFTER INSERT TRIGGER :

```
DELIMITER $$
CREATE TRIGGER simple_after_insert
AFTER INSERT ON Orders
FOR EACH ROW
BEGIN
  UPDATE Customers
  SET balance = balance + NEW.amount
  WHERE customer_id = NEW.customer_id;
END$$
DELIMITER ;
```

4.] Simple AFTER DELETE TRIGGER :

```
DELIMITER $$
CREATE TRIGGER simple_after_delete
AFTER DELETE ON Orders
FOR EACH ROW
BEGIN
  UPDATE Customers
  SET balance = balance - OLD.amount
  WHERE customer_id = OLD.customer_id;
END$$
DELIMITER ;
```

5.] Simple BEFORE UPDATE TRIGGER :

```
DELIMITER $$
CREATE TRIGGER simple_before_update
BEFORE UPDATE ON Orders
FOR EACH ROW
BEGIN
  IF NEW.amount = 0 THEN
    SIGNAL SQLSTATE '45000';
  END IF;
END$$
DELIMITER ;
```

6.] OUTPUT:

Insert and Update values in Order:

```
INSERT INTO Orders VALUES (1, 101, 500);
```

```
UPDATE Orders
  SET amount = 800
  WHERE order_id = 1;
```

```
mysql> SELECT*FROM Orders;
+-----+-----+-----+
| order_id | customer_id | amount |
+-----+-----+-----+
|          1 |          101 | 800.00 |
+-----+-----+-----+
1 row in set (0.00 sec)
```

```
INSERT INTO Orders VALUES (2, 101, 500);
```

```
mysql> SELECT*FROM Orders;
+-----+-----+-----+
| order_id | customer_id | amount |
+-----+-----+-----+
|          1 |          101 | 800.00 |
|          2 |          101 | 500.00 |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
UPDATE Orders
  SET amount = 1000
  WHERE order_id = 2;
```

```
mysql> SELECT*FROM Orders;
+-----+-----+-----+
| order_id | customer_id | amount |
+-----+-----+-----+
|          1 |          101 | 800.00 |
|          2 |          101 | 1000.00 |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

7.] Trigger Output:

```
INSERT INTO Orders VALUES (10, 101, -500);
```

```
mysql> INSERT INTO Orders VALUES (10, 101, -500);
ERROR 1644 (45000): order amount must be greater than zero
```

```
UPDATE Orders
  SET amount = 0
  WHERE order_id = 2;
```

```
mysql> UPDATE Orders
      -> SET amount = 0
      -> WHERE order_id = 2;
ERROR 1644 (45000): Unhandled user-defined exception condition
```

```
DELETE FROM Orders
WHERE order_id = 1;
```

```
mysql> SELECT*FROM Orders;
+-----+-----+-----+
| order_id | customer_id | amount |
+-----+-----+-----+
|          2 |          101 | 1000.00 |
+-----+-----+-----+
1 row in set (0.00 sec)
```